

Summary

Data analyst with deep experience in Python and Pandas. I work with messy, real-world datasets to figure out what the numbers actually mean, whether that's cancellation patterns in hotel bookings, default risk in financial loans, cost burdens in healthcare, loyalty segments in banking, profit margins in e-commerce, or revenue trends in sales. The work covers data cleaning, EDA, statistical analysis, KPI tracking, and building dashboards people can actually use.

Professional Experience

Data Analyst, Codiora House (Private) Limited

06/2026 – 08/2026 | Remote

- Built interactive dashboards using Streamlit and Plotly to track portfolio and business performance metrics.
- Cleaned and restructured raw datasets with Python, Pandas, and NumPy, then ran exploratory analysis to identify patterns in customer behavior and surface relevant business insights.
- Designed KPI dashboards and trend reports that informed executive decisions directly.

Projects

E-Commerce Sales Analysis

02/2026 – 03/2026

- Analyzed 1,000+ orders across categories, regions, and customer segments to uncover revenue and profit patterns.
- Identified Electronics as the highest-sales category and Grocery as the highest profit-margin category.
- Investigated discount impact, found heavy discounting reduces profit by -24% without proportional volume gain.
- Segmented customers (Consumer, Corporate, Home Office) to compare sales, profit, and average order value.

Hotel Booking Analysis | Cancellation & Revenue Insights

03/2026 – 03/2026

- Cleaned and preprocessed a dataset of 119,390 hotel bookings, removing 31,994 duplicates and handling missing values.
- Identified an overall cancellation rate of 27.49%, with City Hotels (30.04%) significantly higher than Resort Hotels (23.48%).
- Found that Non-Refundable deposit bookings had a 94.70% cancellation rate, a key business risk area.
- Discovered that bookings made 90+ days in advance carry the highest cancellation risk due to extended lead time.

Financial Loan Risk & Performance Analysis

04/2026 – 05/2026

- Analyzed a portfolio of 38,576 loans totaling \$435.76M in lending with a 12.05% average interest rate.
- Calculated an overall default (charge-off) rate of 13.82%, broken down by loan grade and sub-grade.
- Found Debt-to-Income ratio is a strong risk predictor up to 25% DTI, after which stricter approval criteria reduce risk.
- Examined the impact of home ownership status and income verification on default probability.

Consumer Electronics Sales Analysis

05/2026 – 05/2026

- Identified total annual revenue of \$34.49M, with December as peak month and January as lowest.
- Found that the top 5 products contributed ~66% of total revenue, flagging revenue concentration risk.
- Found peak buying hours at 11AM–8PM with a 7PM spike, identified as the optimal window for ads and promotions.
- Uncovered product bundling opportunity: customers frequently purchased phones alongside charging accessories

Patient Healthcare Analysis

05/2026 – 06/2026

- Analyzed patient records covering demographics, disease severity, treatment costs, and hospital utilization.
- Found total medicine expenditure of \$9.22M across all patients, with Asthma as the most common diagnosis.
- Identified high-repeat-visit patients (5–7 visits/year) predominantly linked to Diabetes, Hypertension, and Coronary Artery Disease.
- Delivered insights to support cost management, staffing decisions, and improved patient care outcomes.

Bank Customer Analysis

06/2026 – 07/2026

- Analyzed customer demographics and loyalty tiers, found 98% hold multiple products, confirming strong cross-selling success.
- Identified Jade loyalty customers as the highest-value segment with the largest average deposits and loan balances.
- Determined that customers aged 36–65 manage the largest share of total bank assets.
- Provided data-driven recommendations to increase customer value and improve product penetration strategies.

Skills

Programming Python, C++	Data Analysis Libraries Pandas, Numpy	Data Visualization Matplotlib, Seaborn, Plotly	Tools & Software Excel, VS Code
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Education

BS Data Science, University Of Layyah

08/2025 – Present | Layyah, Pakistan

Certificates

Python for Data Science & Machine Learning

Languages

English | Urdu